



**S. B. JAIN INSTITUTE OF
TECHNOLOGY, MANAGEMENT &
RESEARCH, NAGPUR.**

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation Predicate Form

Dog is an Animal isa(dog, animal)

Bird can Fly can(bird, fly)

Dog has Tail has(dog, tail)

Inheritance rule:

Artificial Intelligence Lab (N-PCCCS601P)

If

isa(X, Y) and has(Y, Z)

Then

has(X, Z)

Semantic Network Example

Nodes:

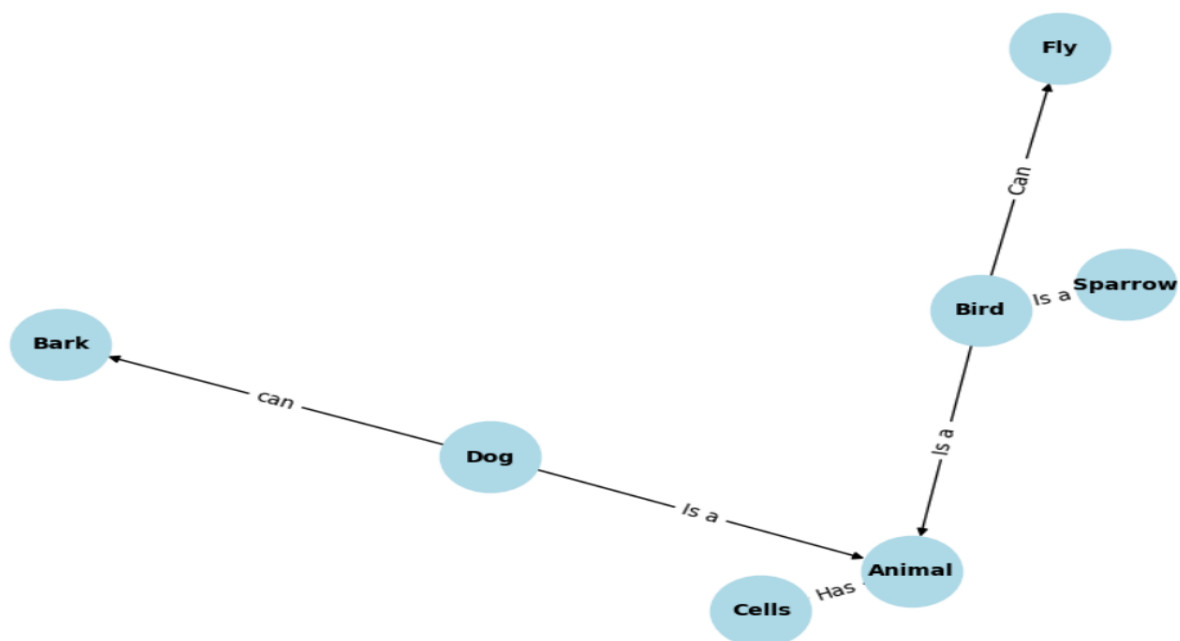
- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:

Semantic Network Graph



PROGRAM CODE:

```
import networkx as nx
import matplotlib.pyplot as plt

class SemanticNetwork:
    def __init__(self):
        self.network = []

    def add_fact(self, subject, relation, obj):
        self.network.append((subject, relation, obj))

    def display_network(self):
        if not self.network:
            print("No facts in the network.")
        else:
            print("\nCurrent Semantic Network:")
            for fact in self.network:
                print(f'{fact[0]} {fact[1]} {fact[2]}')

    def draw_graph(self):
        if not self.network:
            print("No facts to display in graph.")
            return

        G = nx.DiGraph()

        # Add edges with labels
        for subject, relation, obj in self.network:
            G.add_edge(subject, obj, label=relation)

        pos = nx.spring_layout(G) # layout for better visualization

        plt.figure(figsize=(8, 6))

        # Draw nodes and edges
        nx.draw(G, pos, with_labels=True, node_color='lightblue',
                node_size=2000, font_size=10, font_weight='bold')

        # Draw edge labels (relations)
        edge_labels = nx.get_edge_attributes(G, 'label')
        nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels)

        plt.title("Semantic Network Graph")
        plt.show()

def main():
    semantic_network = SemanticNetwork()
```

```
while True:
    print("\nMenu:")
    print("1. Add a new fact")
    print("2. Display all facts")
    print("3. Show graph")
    print("4. Exit")

    choice = input("Enter your choice (1-4): ")

    if choice == '1':
        try:
            num_facts = int(input("How many facts do you want to enter? "))
        except ValueError:
            print("Please enter a valid number.")
            continue

        for i in range(num_facts):
            print(f"\nFact {i + 1}:")
            subject = input("Enter subject: ")
            relation = input("Enter relation: ")
            obj = input("Enter object: ")
            semantic_network.add_fact(subject, relation, obj)

    elif choice == '2':
        semantic_network.display_network()

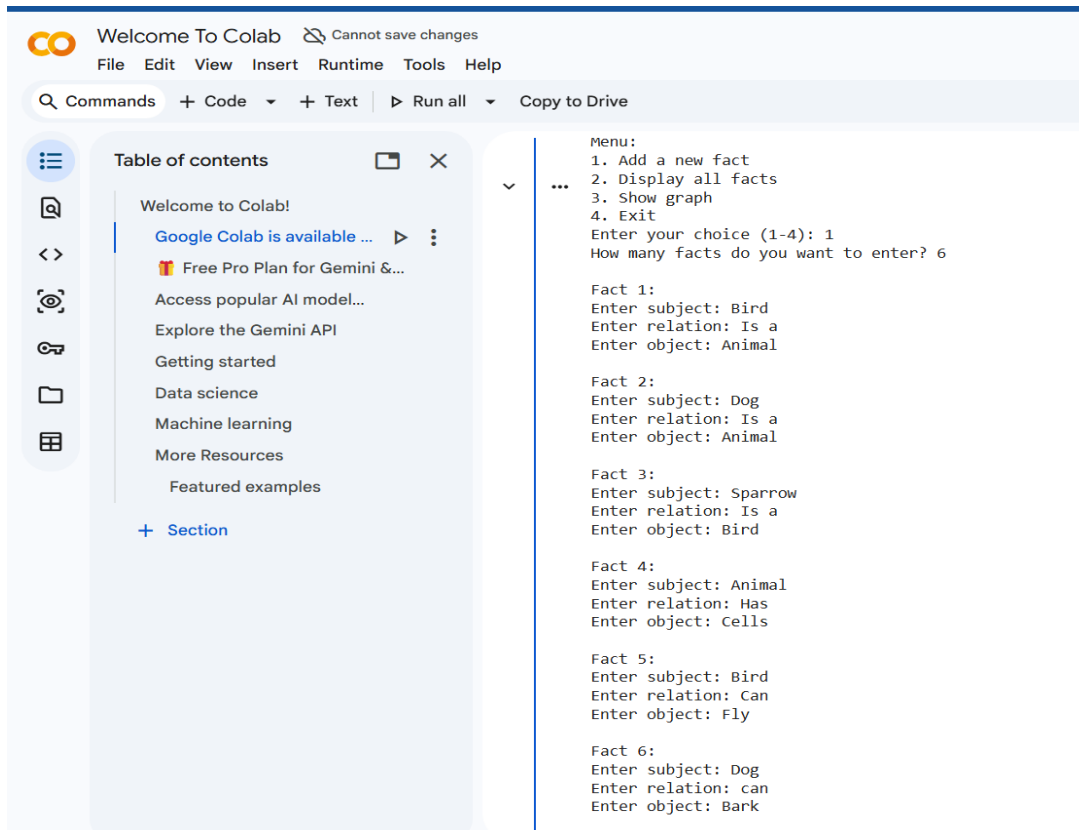
    elif choice == '3':
        semantic_network.draw_graph()

    elif choice == '4':
        print("Exiting the program.")
        break

    else:
        print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```

INPUT & OUTPUT:



Google Colab interface showing the initial state. The table of contents on the left includes: Welcome to Colab!, Google Colab is available..., Free Pro Plan for Gemini &..., Access popular AI model..., Explore the Gemini API, Getting started, Data science, Machine learning, More Resources, and Featured examples. The main editor area displays a menu with the following options:

```

Menu:
1. Add a new fact
2. Display all facts
3. Show graph
4. Exit
Enter your choice (1-4): 1
How many facts do you want to enter? 6

Fact 1:
Enter subject: Bird
Enter relation: Is a
Enter object: Animal

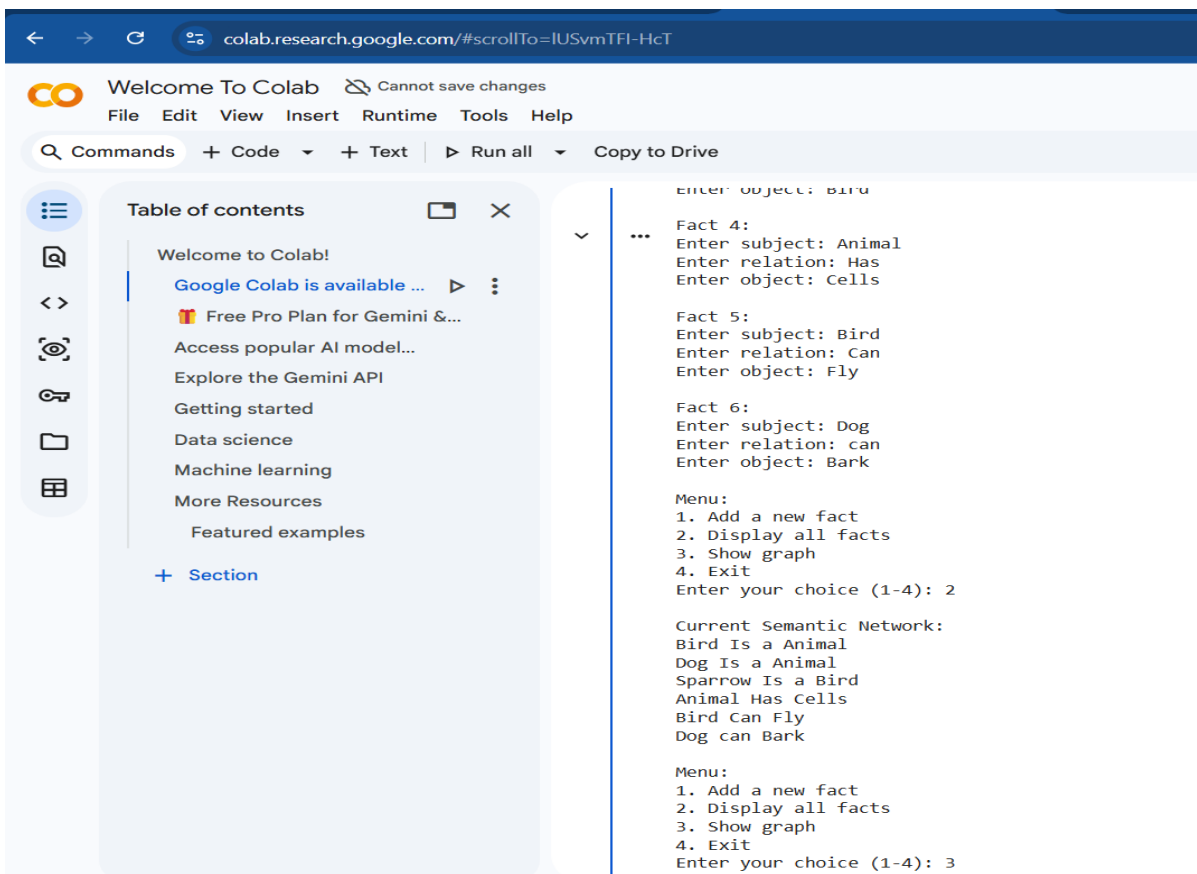
Fact 2:
Enter subject: Dog
Enter relation: Is a
Enter object: Animal

Fact 3:
Enter subject: Sparrow
Enter relation: Is a
Enter object: Bird

Fact 4:
Enter subject: Animal
Enter relation: Has
Enter object: Cells

Fact 5:
Enter subject: Bird
Enter relation: Can
Enter object: Fly

Fact 6:
Enter subject: Dog
Enter relation: can
Enter object: Bark
  
```



Google Colab interface showing the state after processing the input facts. The table of contents on the left remains the same. The main editor area displays the following output:

```

Enter object: Bird

...

Fact 4:
Enter subject: Animal
Enter relation: Has
Enter object: Cells

Fact 5:
Enter subject: Bird
Enter relation: Can
Enter object: Fly

Fact 6:
Enter subject: Dog
Enter relation: can
Enter object: Bark

Menu:
1. Add a new fact
2. Display all facts
3. Show graph
4. Exit
Enter your choice (1-4): 2

Current Semantic Network:
Bird Is a Animal
Dog Is a Animal
Sparrow Is a Bird
Animal Has Cells
Bird Can Fly
Dog can Bark

Menu:
1. Add a new fact
2. Display all facts
3. Show graph
4. Exit
Enter your choice (1-4): 3
  
```

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a semantic network?

2. What is ISA relationship?

3. What is inheritance in knowledge representation?

4. How is predicate logic used here?

5. What are advantages of semantic networks?

REFERENCES:

- <https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial-intelligence/>
- <https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/>
- <https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/>